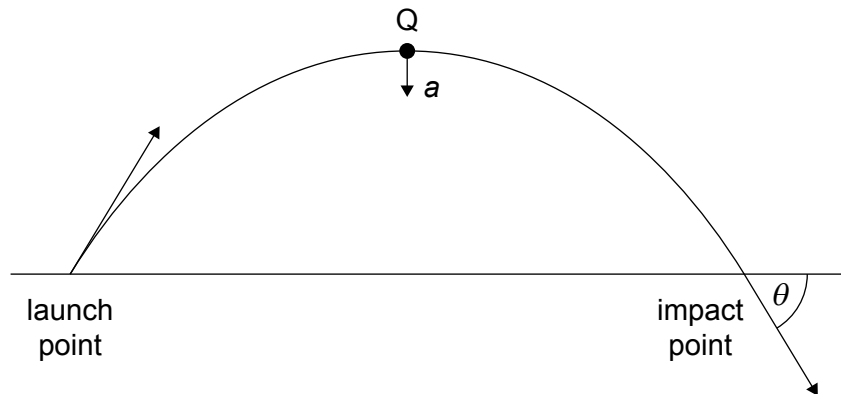


PAPER 1 / QUESTION 32

29 April 2025 / TZ2 (file label) / HL Paper 1A / original question 3 / 2225-6713

32. The diagram shows the path of a ball in the absence of air resistance. Q is the highest point of the ball's trajectory and a is the vertical acceleration at Q. At impact the velocity makes an angle θ to the horizontal.



Three statements about the actual motion of the ball when there is air resistance are:

- I. Q is lower.
- II. a remains the same.
- III. θ increases.

Which statements are correct?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III